

MAGIC3D Electromagnetic FDTD-PIC Code Dense Plasma Model Benchmark

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Abstract: *The MAGIC3D EM FDTD PIC plasma code is compared to MAGIC2D and LSP 3D for dense plasma air ionization test problems. Agreement of maximum responses within one percent is demonstrated.*

Keywords: MAGIC; EM FDTD PIC code; dense plasma; pinched electron beam; LSP

Background

The MAGIC2D electromagnetic (EM) finite difference-time domain (FDTD) particle-in-cell (PIC) code [1] contains an air ionization, dense plasma physics model, similar to that in the LSP (Large Scale Plasma) code [2]. LSP has been applied to a number of EM plasma investigations in recent years [3]-[5]. MAGIC users also model gas plasmas, so comparison of results with the more recent (2005) LSP model results was considered prudent and instructive. This exercise was recently accomplished for MAGIC2D [6].

Introduction

MAGIC3D FDTD PIC code [1] EM responses using the dense plasma model are compared with MAGIC2D and LSP 3D [2] responses in this work. Excellent agreement between the tools is achieved when using simple geometric representations of generic electron beam chamber experiments. Validation of the models, in toto, is demonstrated. The desired goal of this comparison is the demonstration of agreement of key responses within approximately one-percent (1%). The essential physics equivalence of the respective plasma models is, of course, required to meet this objective.

The approach taken is to first compare computations on a simple, small, empty, square duct; and then proceed to a more complex system, representative of a large area electron beam experiment. These configurations facilitate highly resolved investigations of the plasma cell behavior. Both problem configurations were first run for the vacuum condition to ensure code agreement in the simplest phenomenological case. The square duct configuration emphasizes the electric field, which is the primary driver of the plasma fluid model. Critical algorithm parameter settings, which facilitate agreement between the codes, are identified.

Air Plasma Model Parameters

Air ionization is caused by the impact of both primary and secondary emission electrons with neutral air molecules. Primaries are energetic electron beam particles and secondaries are low energy ionization products which are also capable of ionizing neutrals. Processes of electron attachment to neutral species, recombination with ions, and neutralization of ions have been compared, but these results are not included here. The LSP generic air plasma model is employed with the excluded rates set to zero.

Air ionization model formulae are examined and the most up-to-date chosen in order to ensure consistency of the models for these detailed code comparisons. SI (MKS) units are used in the MAGIC code and in this paper (with a few standard exceptions): Pressure P is in atmospheres and electron energy E_e and energy equivalent temperature T_e , in electron-volts (eV). $N=2.6868 \times 10^{25} P$ is the gas molecule number density ($\#/m^3$) using Loschmidt's number at STP. The plasma model is driven by E/P , where E is the electric field only (inductive terms ignored as unimportant).

The primary electron neutral ionization cross section is derived from the electron stopping power dE/dx of the medium,

$$\sigma_{ion} = A_0 dE/dx / (N W_{gas} c)$$

$$dE/dx = A_1 N Z_{gas} Z_{eff} [\ln(A_2 I_{sq} E_e \gamma^2 \beta^2) + 1 - \beta^2] / \beta^2$$

where: W_{gas} is the energy needed per created pair (34.0 eV); γ is the relativistic factor; β the ratio of the electron velocity to c (the vacuum speed of light); Z_{gas} is the number of bound electrons per molecule (14 used here); and $Z_{eff} = 1$. $I_{sq} = (I_{av})^{-2}$ where: $I_{av} = 80 \text{ eV}/m_e c^2$ is the average electron ionization potential ($m_e c^2 = 5.11 \times 10^5$); $A_0 = 1.54 \times 10^{10}$; $A_1 = 4.98 \times 10^{-25}$ and $A_2 = 0.35355$ are material-dependent constants. The expression dE/dx has an embedded factor of $m_e c^2$.

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The resultant formula for σ_{ion} conservatively replicates National Institute of Standards and Technology data for dry air over the energy range utilized here.

The avalanche coefficient is:

$$\alpha(E/P) = 100c_1 f_{ion} c_1 (E/P)^6 / (1 + c_2 E/P + c_3 (E/P)^4 + c_4 (E/P)^6) / (1.0 + T_e) / (4.0 * W_{ion})$$

where E is the electric field. Constant values are:

$c_1 = 7.84 \times 10^4$; $c_2 = 26.67$; $c_3 = 2242$;

$c_4 = 691.6$; and f_{ion} is the molecule fraction which is not ionized ($f_{ion} \sim 1.0$ in this work).

The electron mobility is $\mu_e (m^2/V-s)$:

$\mu_e (|E|, P, n_e) = e / (100 m_e \nu_c)$, where

$\nu = \nu_{en} (|E|, P) + \nu_{ei} (|E|, n_e)$.

The terms ν_{en} and ν_{ei} are the electron collision rates with neutrals and ions, respectively. The argument n_e is the secondary electron number density ($\#/m^3$) which is computed with the standard rate equation [1].

$\nu_{en} (|E|, P) = 64.0 P (T_e)^{1/2}$ for $T_e \geq 4.0$

$\nu_{en} (|E|, P) = P (4.98733 + T_e (82.3167 + T_e (-20.7957 + 1.97517 T_e)))$ for $T_e < 4.0$

$\nu_{ei} (|E|, n_e) = 4.86 \times 10^{-17} f_e (10^{-6} n_e) (T_e)^{-1.5} \ln(1.0 + 2.39 \times 10^{20} (T_e)^3 / (10^{-6} n_e))$

where $f_e = \text{MAX}(1, 0.5 n_e / N)$ is the fractional ionization and the factor of 1/2 is for diatomic molecules.

Using $\phi = \sqrt{10^{-5} E/P / 511.0}$, the electron temperature from a least squares fit to a cubic spline is:

$T_e = a_1(1.0 + a_2 \phi (1.0 + a_3 \phi (1.0 + a_4 \phi (1.0 - a_5 \phi (1.0 - a_6 \phi (1.0 - a_7 \phi (1.0 - a_8 \phi))))))$ for $\phi < 0.716$ and

$T_e = b_1 (1 + b_2 \phi)$ for $\phi \geq 0.716$

where constant values are $a_1 = 0.155$;

$a_2 = 8.6277$, $a_3 = 14.944$, $a_4 = 18.948$;

$a_5 = 5.6968$, $a_6 = 2.1216$, $a_7 = 0.92303$;

$a_8 = 0.33829$; $b_1 = 0.505$, and $b_2 = 35.4$.

The electron range of 3.8 m is significantly longer than the simple geometry employed here, but is comparable to the pinch geometry case.

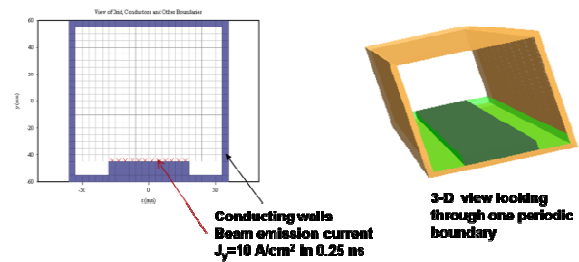
Revisions to the ionization model were made subsequent to this investigation. An improved set of ionization cross sections and stopping powers were implemented. These employ the Beta formula for high energy, and are validated for air against the National Institutes of Standards and Technology (NIST) data base and codes. Low energy cross sections and stopping power use an internal table for a few gases, and then match the Beta stopping power prediction with the table, renormalizing data above 300 eV.

The ion charge is captured in a reservoir array allowing the generation of a fluid momentum transfer field associated with the implied ion-neutral collisions. The model is designed for relatively dense neutral gas and high energy primary electrons, but is expected to perform at modest and low energies, as well.

Simple Geometry Electron Beam Test

The small scale test problem is a square duct of effectively infinite length (the 3D model is 1 meter long with periodic boundaries at each end, Fig. 1). The interior cross section is 11 cm by 11 cm. The system is modeled in Cartesian coordinates with 0.5 cm zones. Electrons are emitted uniformly from the raised cathode at $y = -4.5$ cm with 1-MeV energy. The emission current density rises to 10 A/cm² in 0.25 ns and then remains constant. The cavity is filled with air at 1 atmosphere. The results of this simple case support the more complex pinched electron beam analysis.

Agreement of the responses is obtained when compatible algorithm settings are used, so the codes solve the same fundamental problem description. The codes are both run in standard momentum-conserving explicit mode. Kinetic ionization requires random emission for comparable resultant code particle trajectories for the nominal numerical grids employed here.



Long duct (100 cm) with periodic boundaries at ends
11x11 cm internal cross section
3-D Cartesian grid, $\Delta x, \Delta y, \Delta z = 5$ mm

Fig. 1. Duct test problem facilitating microscopic investigation of plasma cell behavior

The results for the electric field at the midpoint of the second zone above the center of the cathode are shown in Fig. 2. Excellent agreement is seen for this primary

driver of the air fluid models with strong quenching of the electric fields by the plasma (compared with the vacuum case). Attention can be directed with confidence to the more complex model of interest.

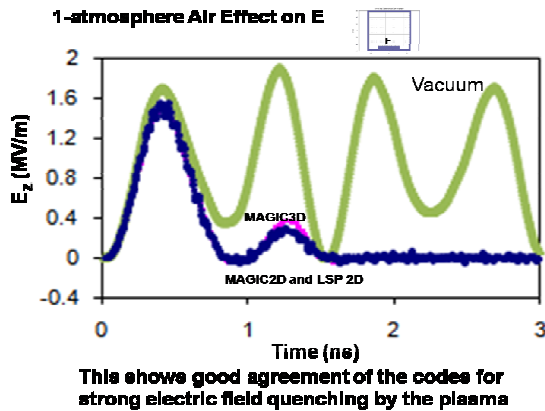


Fig. 2. Peak electric field driver of plasma model in duct.

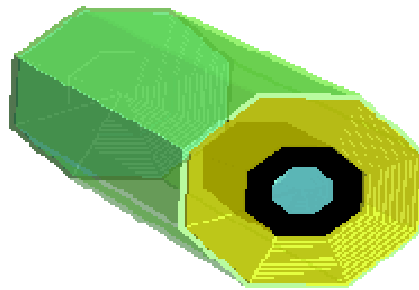
Pinched Electron Beam Test

The pinched electron beam test geometry is a cylindrical cavity containing air at 1 atmosphere represented in cylindrical coordinates (Fig. 3). The lower drawing shows the cross section in the r - z plane which is rotated about the symmetry axis in the codes to form the 1.2 m diameter by 4.5 m length conducting chamber. The upper drawing shows the computer 3D representation for the modest 45-degree polar grid employed here in 3D.

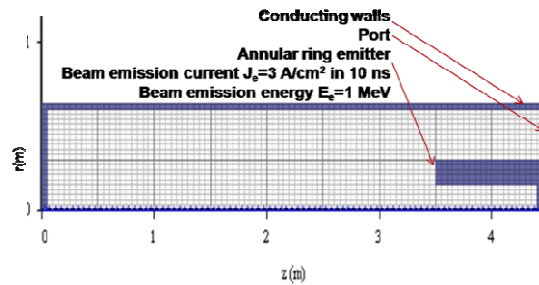
A port is used at the top right (Fig. 3.b) to represent the impedance of the cathode assembly relative to the chamber. 1-MeV electrons are emitted uniformly from the end of the circular cathode over an annulus of interior radius 15 cm, and exterior radius, 30 cm. The current density rises to 3 A/cm² in 10 ns, then remains constant. The initial magnetic field is zero.

In Fig. 4, the electron positions are shown at time 25 ns for the nominal case. The modest differences in the initial pinch location are a consequence of slightly different numerical treatments in the codes (MAGIC has zone centered conductivity whereas LSP has zone edge). These differences decrease substantially with decreased spatial cell size (by a factor of 3 to 4) based on the 2D results shown at 14 ns for 4x reduced zone sizes in Fig. 5. The magnetic field contours at 30 ns are shown in Fig. 6. Substantial agreement of both the positions and resulting fields are seen. Both codes give a maximum pinched magnetic field of ~0.018 tesla at radius ~3 cm, 1.8 meters from the cathode at time 30 ns.

a. 3D view looking in direction of beam:



b. Side view looking in across beam:



3-D Cylindrical grid
Nominal case cell sizes: $\Delta z = 1.25$ cm, $\Delta r = 0.75$ cm, $\Delta \phi = 45$ degrees
 $B_{static} = 0$

Fig. 3. Test problem geometry for pinched beam comparison.

Summary and Conclusions

The MAGIC3D EM FDTD PIC code has been compared with MAGIC2D and LSP responses on air ionization test problems using simple geometric representations of generic electron beam chamber experiments. Nominally, a one-percent agreement among important maximum responses has been obtained. The pinched beam response demonstrates significant agreement of the FDTD PIC dense plasma models, in toto. The zone-by-zone travelling nature of the air plasma breakdown of the electric fields causes the transition from electric to magnetic field determination of electron trajectories, thus stressing every core aspect of the codes.

Careful selection of the algorithm settings is required due largely to the size of the finite difference grids employed combined with some modest variations in the details of the code parameter settings. Statistical variations in the results are evident, but the impact of these can be made inconsequential with randomized emission, decreased time step, and increased particle numbers.

Based on results of detailed comparisons, such as the examples presented here, MAGIC code users can be

confident they will obtain responses in close agreement with LSP code analysis for dense plasmas. MAGIC versions 2.1.9 and greater have both the 3D plasma model and the upgraded air model rates consistent with NIST data. Additionally, MAGIC provides input of arbitrary gas chemistry functions by the user employing the MAGIC Control Language MCL. This enables the convenient and accurate representation of more complex gas mixtures producing dense plasmas.

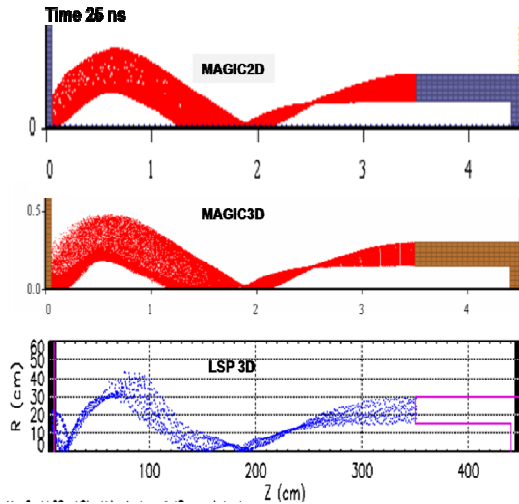


Fig. 4. Particles vs. radial (vertical) and axial position at time 25 ns from MAGIC (top) and LSP for the nominal grid case.

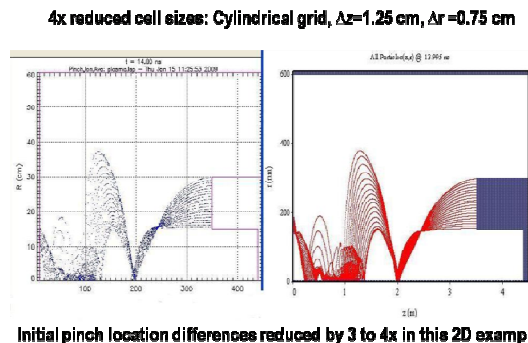


Fig. 5. Particles vs. radial (vertical) and axial position at time 14 ns from MAGIC (right) and LSP for 4x decreased cell sizes ($\Delta z = 1.25$ cm, $\Delta r = 0.75$ cm).

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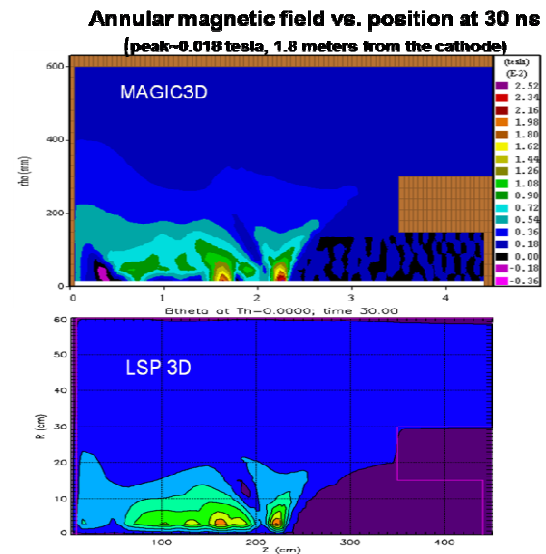


Fig. 6. Annular magnetic field vs. radial (vertical) and axial position at time 30 ns from MAGIC3D (top) and LSP 3D. (1 gauss = 10^4 tesla).

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